

MTH 204: Lecture 14

Critical point classification

Consider the linear system: $Y' = AY = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} Y$

The characteristic polynomial of A :

$$\det(A - \lambda I) = (a_{11} - \lambda)(a_{22} - \lambda) - a_{12}a_{21}$$

$$= \lambda^2 - (a_{11} + a_{22})\lambda + (a_{11}a_{22} - a_{12}a_{21})$$

$$= \lambda^2 - \operatorname{Tr}(A)\lambda + \det(A)$$

Let us define $\Delta = \operatorname{Tr}(A)^2 - 4\{\det(A)\}$

The eigen values of A are the roots of the equation $\det(A - \lambda I) = 0$

Hence the eigen values are $\lambda_1, \lambda_2 = \frac{\operatorname{Tr}(A) \pm \sqrt{\Delta}}{2}$

Note that $\det(A - \lambda I) = (\lambda - \lambda_1)(\lambda - \lambda_2)$

Therefore $\lambda_1 + \lambda_2 = \operatorname{Tr}(A) = p$ (say)

and $\lambda_1 \lambda_2 = \det(A) = q$ (say)

and $\Delta = (\lambda_1 + \lambda_2)^2 - 4\lambda_1 \lambda_2 = (\lambda_1 - \lambda_2)^2$

Based on the sign of $\text{Tr}(A)$, $\det(A)$ and Δ we can classify the critical points.

Criteria for Critical points:

Type	$p = \text{Tr}(A)$ $= \lambda_1 + \lambda_2$	$q = \det(A)$ $= \lambda_1 \lambda_2$	$\Delta = (\lambda_1 - \lambda_2)^2$	Comments
Node		> 0	> 0	Real, same sign
Saddle point		< 0		Real, opposite sign
Center	$= 0$	> 0	< 0	Pure imaginary
Spiral point	$\neq 0$		< 0	Complex, not pure imaginary

Stability:

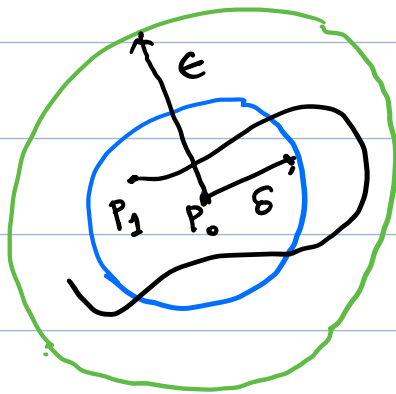
Stability means, roughly speaking, that a small change (a small disturbance) of a physical system at some instant changes the behavior of the system only slightly all all future times t .

Stable critical point:

A critical point P_0 is stable if all trajectories

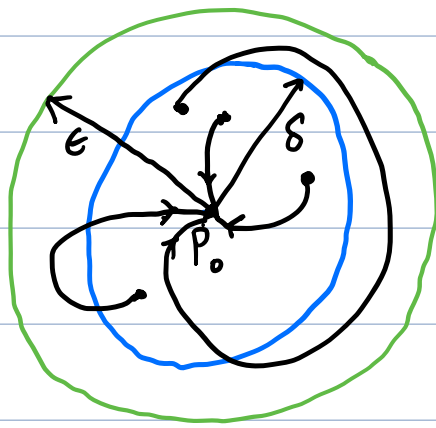
of the ODE that at some instant are close to P_0 remain close to P_0 at all future times;
 More precisely: If every disk D_ϵ of radius ϵ with center P_0 there is a disk D_δ of radius δ with center P_0 such that every trajectory of the ODE that has a point P_1 (corresponding to $t = t_1$, say) in D_δ has all its points corresponding to $t \geq t_1$ in D_ϵ .

If a critical point is not stable, it is unstable.



Asymptotically stable critical point:

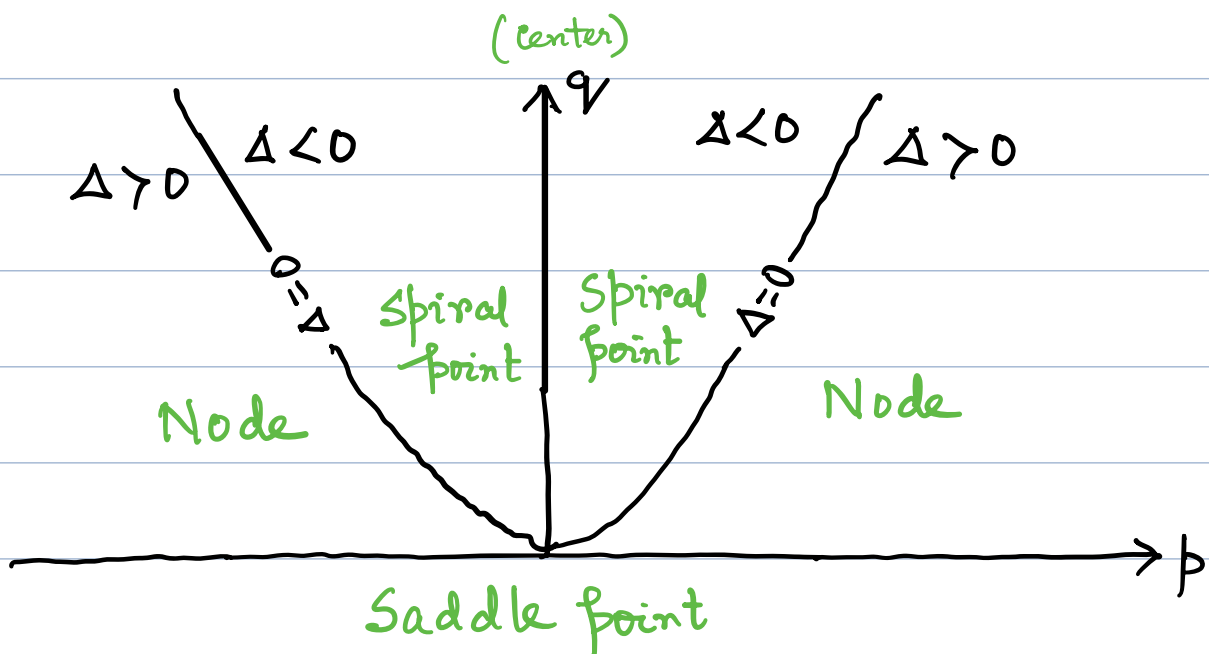
A critical point P_0 is asymptotically stable (stable and attractive) if P_0 is stable and every trajectory that has a point in D_δ approaches P_0 as $t \rightarrow \infty$.



(stable and attractive critical point P_0)

Criteria for Critical points:

Type	$p = \text{Tr}[A]$ $= \lambda_1 + \lambda_2$	$q = \det A$ $= \lambda_1 \lambda_2$
Asymptotically stable	< 0	> 0
Stable	≤ 0	> 0
Unstable	$p > 0$ or $q < 0$	



Stability chart of the system

Stable and attractive: The second quadrant
without the q -axis.

Stability also on the positive q -axis

(which corresponds to centers)

Unstable: The first quadrant excluding the
 q -axis and the region below
the p -axis

Classification of Critical points :

(1) Eigen values are real , distinct

(a) Both positive : improper node , unstable

(b) Both negative : improper node , stable , asymptotically stable

(c) One positive , another negative : Saddle point
unstable

(2) Eigen values are real , equal :

(a) Positive , eigen space 1-dimensional : degenerate node
unstable

(b) Positive , eigen space 2-dimensional : proper node
(star) unstable (non degenerate)

(c) Negative , eigen space 1-dimensional : degenerate
node , stable and asymptotically stable

(d) Negative , eigen space 2-dimensional : proper
node (star) ; stable and asymptotically stable
(non degenerate)

(3) Eigen values are Purely imaginary (real part zero):
center , stable (not asymptotically stable)

(4) Eigen values are complex (but not purely imaginary):

(a) Real part positive: Spiral point, unstable

(b) Real part negative: Spiral point, stable
and asymptotically stable

Example: Free motions of a mass on a spring:

$$m y'' + c y' + k y = 0 \quad (c > 0, k > 0)$$

$$\Rightarrow y'' = -\frac{c}{m} y' - \frac{k}{m} y$$

Let $y_1 = y$ and $y_2 = y'$

Then we have an equivalent system of equations

$$\left. \begin{aligned} y_1' &= y_2 \\ y_2' &= -\frac{k}{m} y_1 - \frac{c}{m} y_2 \end{aligned} \right\}$$

$$\Rightarrow \begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$\Rightarrow Y' = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{pmatrix} Y = A Y \text{ (say)}$$

$$\text{where } A = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{pmatrix}$$

$$\begin{aligned}\text{Now } \det(A - \lambda I) &= (-\lambda)\left(-\lambda - \frac{c}{m}\right) + \frac{k}{m} \\ &= \lambda^2 + \frac{c}{m}\lambda + \frac{k}{m}\end{aligned}$$

If λ_1 and λ_2 are two eigen values, then

$$p = \lambda_1 + \lambda_2 = \text{tr}(A) = -\frac{c}{m}$$

$$q = \lambda_1 \lambda_2 = \det(A) = \frac{k}{m}$$

$$\text{And } \Delta = p^2 - 4q = \left(\frac{c}{m}\right)^2 - 4\frac{k}{m} = \frac{c^2 - 4mk}{m}$$

Thus we have the following situations:

No damping: $c=0, p=0, q>0$: a center

Underdamping: $c^2 < 4mk, p < 0, q > 0, \Delta < 0$

a stable and attractive spiral point

Critical damping: $c^2 = 4mk, p < 0, q > 0, \Delta = 0$

a stable and attractive node.

Over-damping: $c^2 > 4mk, p < 0, q > 0, \Delta > 0$

a stable and attractive node;

Qualitative Methods for Nonlinear Systems

- Qualitative methods are methods of obtaining qualitative information on solutions without

actually solving a system.

- Useful for systems whose solution by analytic method is difficult or impossible.

For nonlinear system $Y' = f(Y)$ ie.
$$\begin{cases} y_1' = f_1(y_1, \dots, y_n) \\ y_2' = f_2(y_1, \dots, y_n) \\ \vdots \\ y_n' = f_n(y_1, \dots, y_n) \end{cases}$$

we may have several critical points

ie. points $(y_1, y_2, \dots, y_n)$ at which $f_i(y_1, y_2, \dots, y_n) = 0$
for $i = 1, 2, \dots, n$

- If Y_0 is a critical point, we may shift the origin so that Y_0 is centered

$$\tilde{Y} = Y - Y_0$$

$$\tilde{Y}' = f(\tilde{Y} + Y_0) \Rightarrow \tilde{Y}' = \tilde{f}(\tilde{Y})$$

and study the local behavior of the system of ODE around 0.

We will also assume that the critical points are isolated.

First we may need to linearize the ODE system. We will assume that the system is autonomous ie the independent variable t does not occur explicitly.

Linearization of Autonomous Nonlinear System:

The system $\tilde{Y}' = \tilde{f}(\tilde{Y})$ can be linearized as:

$$\tilde{\mathbf{y}}' = \tilde{\mathbf{f}}(\tilde{\mathbf{y}}) \approx A\tilde{\mathbf{y}}$$

where A is the Jacobian of the function $\tilde{\mathbf{f}}$ evaluated at the origin

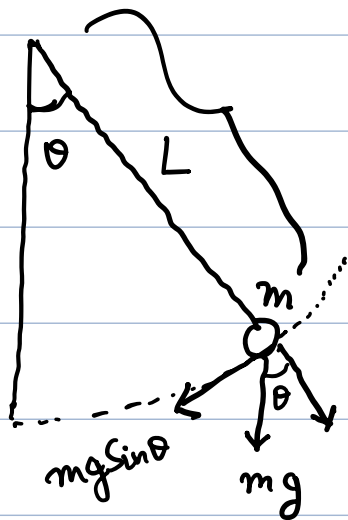
$$A = \begin{pmatrix} \frac{\partial \tilde{f}_1}{\partial \tilde{y}_1} & \frac{\partial \tilde{f}_1}{\partial \tilde{y}_2} & \dots & \frac{\partial \tilde{f}_1}{\partial \tilde{y}_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial \tilde{f}_n}{\partial \tilde{y}_1} & \frac{\partial \tilde{f}_n}{\partial \tilde{y}_2} & \dots & \frac{\partial \tilde{f}_n}{\partial \tilde{y}_n} \end{pmatrix} \bigg|_{\tilde{\mathbf{y}} = \mathbf{0} \rightarrow \begin{pmatrix} \text{zero} \\ \text{vector} \end{pmatrix}}$$

Theorem:

If $\tilde{\mathbf{f}}$ has continuous components and continuous partial derivatives in a neighborhood of the critical point $\mathbf{0}$ and $\det(A) \neq 0$, then the kind and stability of the critical point of the nonlinear system of ODE is the same as those of the linearized system.

Exceptions occur if A has equal or pure imaginary eigenvalues, then the nonlinear system may have the same kind of critical point as the linearized system or a spiral point.

Ex: Free Undamped Pendulum:



Gravity compensates the acceleration of the bob.

$$mL\theta'' + mg\sin\theta = 0$$

$$\Rightarrow \theta'' + k\sin\theta = 0 \text{ where } k = \frac{g}{L}$$

To find the critical points, we convert the equation into a system of ODEs:

$$\left. \begin{array}{l} y_1 = \theta, \quad y_2 = y_1' \\ \text{Then } y_2' + k\sin y_1 = 0 \end{array} \right\}$$

$$\text{Thus } Y' = \begin{pmatrix} y_2 \\ -k\sin y_1 \end{pmatrix}$$

The critical points are $\begin{pmatrix} n\pi \\ 0 \end{pmatrix}$ for any integer n .

Let us look at the critical point $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

The Jacobian of f at $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$$A = \begin{pmatrix} \frac{\partial f_1}{\partial y_1} & \frac{\partial f_1}{\partial y_2} \\ \frac{\partial f_2}{\partial y_1} & \frac{\partial f_2}{\partial y_2} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -k\cos y_1 & 0 \end{pmatrix} \bigg|_{\begin{pmatrix} 0 \\ 0 \end{pmatrix}} = \begin{pmatrix} 0 & 1 \\ -k & 0 \end{pmatrix}$$

$$\text{Tr}(A) = 0, \quad \det A = k > 0$$

So, $y=0$ is a center (always stable)
 (Same is true at all points $\begin{pmatrix} 2n\pi \\ 0 \end{pmatrix}$ since the
 sin function is periodic with period 2π)

Let us now look at the point $\begin{pmatrix} \pi \\ 0 \end{pmatrix}$.

We ~~consider~~ this critical point by using
 the transformation $\tilde{y} = y - \begin{pmatrix} \pi \\ 0 \end{pmatrix}$ (ie $\tilde{y}_1 = y_1 - \pi$
 $\tilde{y}_2 = y_2$)

The system of ODE now becomes:

$$\tilde{y}' = \begin{pmatrix} \tilde{y}_2 \\ -k \sin(\tilde{y}_1 + \pi) \end{pmatrix}$$

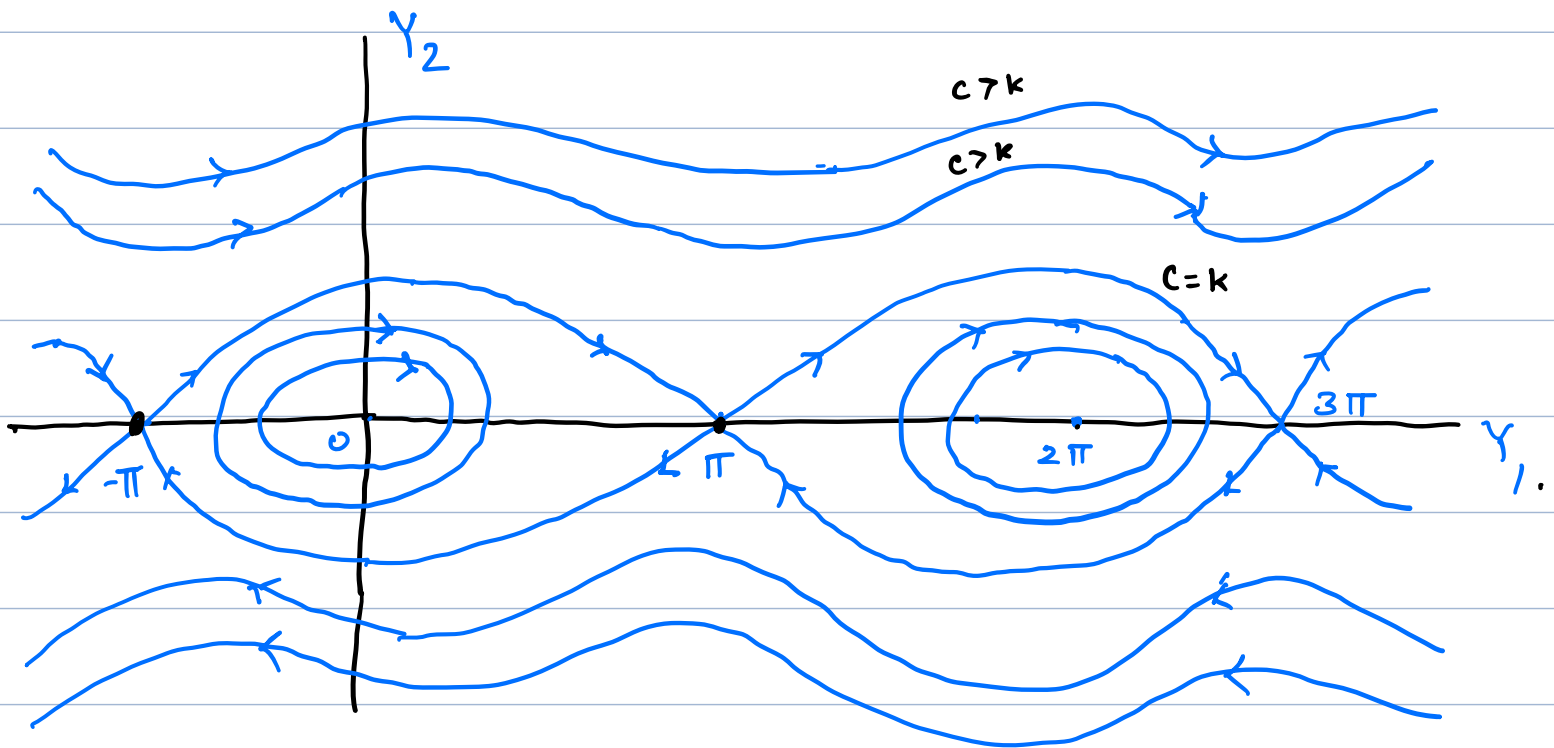
Let us calculate the jacobian of $\tilde{f}$ at $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$$A = \left(\begin{array}{cc} \frac{\partial \tilde{f}_1}{\partial \tilde{y}_1} & \frac{\partial \tilde{f}_1}{\partial \tilde{y}_2} \\ \frac{\partial \tilde{f}_2}{\partial \tilde{y}_1} & \frac{\partial \tilde{f}_2}{\partial \tilde{y}_2} \end{array} \right) \bigg|_{\substack{\tilde{y} = 0 \\ \text{(zero vector)}}} = \left(\begin{array}{cc} 0 & 1 \\ -k \cos(\tilde{y}_1 + \pi) & 0 \end{array} \right) \bigg|_{\tilde{y} = 0} = \begin{pmatrix} 0 & 1 \\ k & 0 \end{pmatrix}$$

Now $\text{Tr}(A) = 0$, $\det(A) = -k < 0$

So, $\gamma = \begin{pmatrix} \pi \\ 0 \end{pmatrix}$ is a saddle point (unstable).

The same is true for all points $\begin{pmatrix} \pi + 2n\pi \\ 0 \end{pmatrix}$ since
sin function is periodic with period 2π



Example: Damped Pendulum:

$$\theta'' + c\theta' + k\sin\theta = 0$$

$$\text{Let } y_1 = \theta, y_2 = y_1'$$

$$\text{Then } y_2' + cy_2 + k\sin(y_1) = 0$$

$$\Rightarrow Y' = \begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} y_2 \\ -k \sin y_1 - c y_2 \end{pmatrix}$$

Critical points are $\begin{pmatrix} n\pi \\ 0 \end{pmatrix}$ for any integer n
 (as in the case of free undamped pendulum)

First consider the critical point $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$:

$$A = \left(\begin{array}{cc} \frac{\partial f_1}{\partial y_1} & \frac{\partial f_1}{\partial y_2} \\ \frac{\partial f_2}{\partial y_1} & \frac{\partial f_2}{\partial y_2} \end{array} \right) \bigg|_{Y=0} = \left(\begin{array}{cc} 0 & 1 \\ -k \cos(y_1) & -c \end{array} \right) \bigg|_{Y=0}$$

$Y=0$
(zero vector)

$$= \begin{pmatrix} 0 & 1 \\ -k & -c \end{pmatrix}$$

$$\text{Tr}(A) = -c < 0, \quad \det(A) = k > 0, \quad \Delta = -c + 4k^2$$

- If $\Delta < 0$, we have a stable and attractive spiral point.
- If $\Delta > 0$, then it is a stable and attractive node.

Let us now look at the critical point $\begin{pmatrix} \pi \\ 0 \end{pmatrix}$

We confirm it by the transformation:

$$\tilde{Y} = Y - \begin{pmatrix} \pi \\ 0 \end{pmatrix} \quad (\text{ie. } \tilde{y}_1 = y_1 - \pi, \tilde{y}_2 = y_2)$$

Then the ODE becomes

$$\tilde{Y}' = \begin{pmatrix} \tilde{y}_2 & 0 \\ -k \sin(\tilde{y}_1 + \pi) & -c \tilde{y}_2 \end{pmatrix}$$

Now

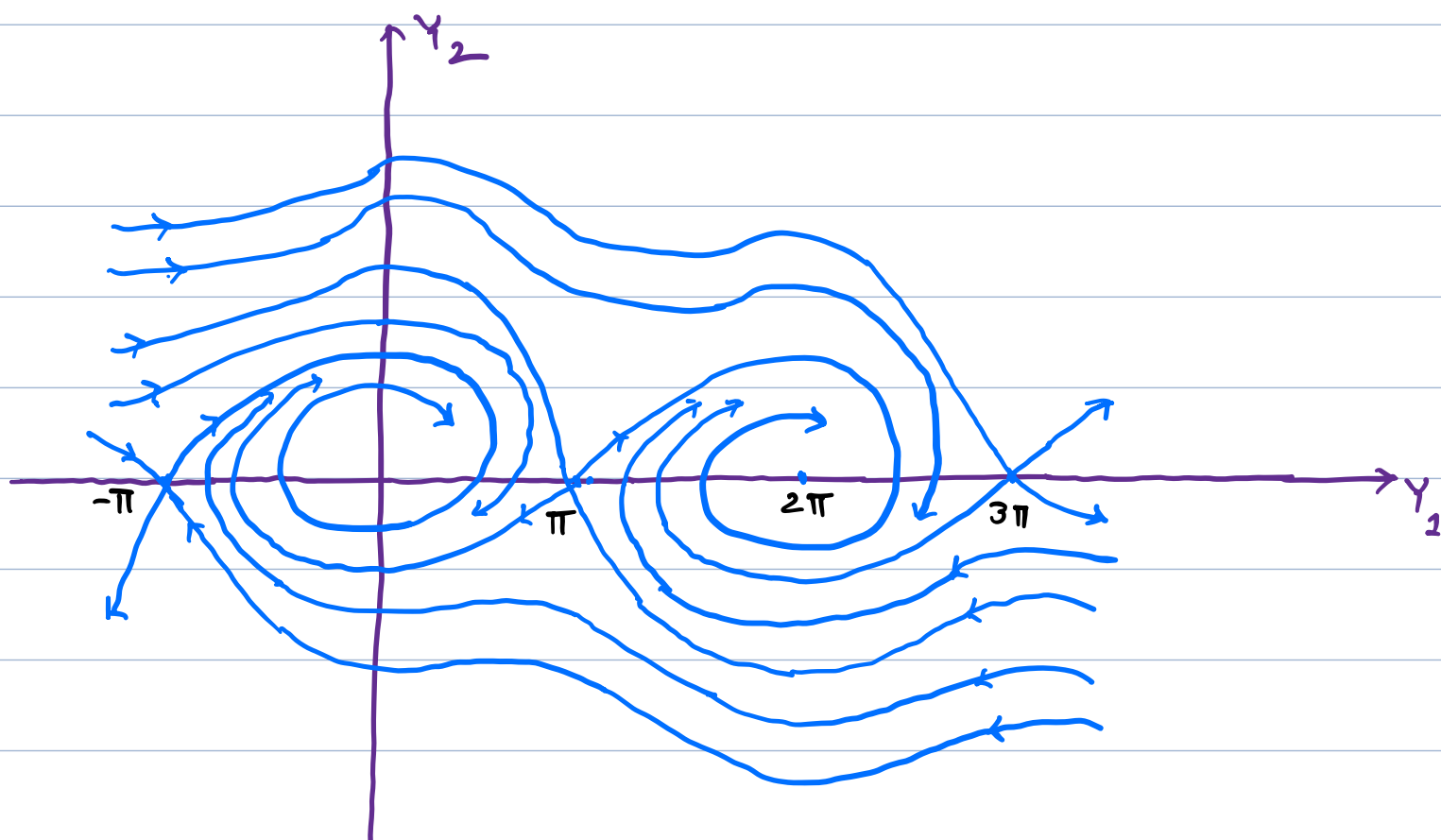
$$A = \left(\begin{array}{cc} \frac{\partial \tilde{f}_1}{\partial \tilde{y}_1} & \frac{\partial \tilde{f}_1}{\partial \tilde{y}_2} \\ \frac{\partial \tilde{f}_2}{\partial \tilde{y}_1} & \frac{\partial \tilde{f}_2}{\partial \tilde{y}_2} \end{array} \right) \bigg|_{\tilde{Y} = \mathbf{0} \text{ (zero vector)}} = \left(\begin{array}{cc} 0 & 1 \\ -k \cos(\tilde{y}_1 + \pi) & -c \end{array} \right) \bigg|_{\tilde{Y} = \mathbf{0}}$$
$$= \begin{pmatrix} 0 & 1 \\ k & -c \end{pmatrix}$$

Note that $\text{Tr}(A) = -c < 0$, $\det A = -k < 0$

So, $Y = \begin{pmatrix} \pi \\ 0 \end{pmatrix}$ is a saddle point (unstable)

The same happens to all points $\begin{pmatrix} \pi + 2n\pi \\ 0 \end{pmatrix}$

Since $\sin$ function is periodic with period 2π .



Ex: Lotka-Volterra Population model:

- Rabbits have unlimited food supply.

Hence, if there were no foxes, their number $y_1(t)$ would grow exponentially, $y_1' = ay_1$

- Actually y_1 is decreased because of the kill by foxes, say at a rate proportional to $y_1 y_2$ where $y_2(t)$ is the number of foxes.

Hence $y_1' = ay_1 - by_1 y_2$ where $a > 0, b > 0$

- If there were no rabbits, then $y_2(t)$ would exponentially decrease to zero: $y_2' = -ly_2$

However y_2 is increased by a rate proportional to the number of encounters between predator and prey; together we have $y_2' = -ly_2 + ky_1 y_2$ where $k > 0$ and $l > 0$

$$\text{Thus } \left. \begin{aligned} y_1' &= ay_1 - by_1 y_2 \\ y_2' &= ky_1 y_2 - ly_2 \end{aligned} \right\}$$

Critical points are solutions of

$$0 = y_1' = y_1(a - by_2)$$

$$0 = y_2' = (ky_1 - l)y_2$$

That is $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} \frac{l}{k} \\ \frac{a}{b} \end{pmatrix}$

First let us look at $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$$A = \left(\begin{array}{cc} \frac{\partial f_1}{\partial y_1} & \frac{\partial f_1}{\partial y_2} \\ \frac{\partial f_2}{\partial y_1} & \frac{\partial f_2}{\partial y_2} \end{array} \right) \bigg|_{y=0} = \left(\begin{array}{cc} a - by_2 & -by_1 \\ ky_2 & ky_1 - l \end{array} \right) \bigg|_{y=0}$$

$y=0$ (zero vector)

$$= \begin{pmatrix} a & 0 \\ 0 & -l \end{pmatrix}$$

Eigen values are $\lambda_1 = a$, $\lambda_2 = -l$

They have different signs & so we have a saddle point.

For the critical point $\left(\frac{l}{k}, \frac{a}{b}\right)$ we use the transformation:

$$\tilde{y} = y - \begin{pmatrix} \frac{l}{k} \\ \frac{a}{b} \end{pmatrix}$$

The system of ODE becomes:

$$\begin{aligned}\tilde{y}' &= \begin{pmatrix} \left(\tilde{y}_1 + \frac{l}{k}\right) \times \left[a - b\left(\tilde{y}_2 + \frac{a}{b}\right)\right] \\ \left[k\left(\tilde{y}_1 + \frac{l}{k}\right) - l\right] \left(\tilde{y}_2 + \frac{a}{b}\right) \end{pmatrix} \\ &= \begin{pmatrix} \left(\tilde{y}_1 + \frac{l}{k}\right) \times (-b\tilde{y}_2) \\ (k\tilde{y}_1) \times \left(\tilde{y}_2 + \frac{a}{b}\right) \end{pmatrix}\end{aligned}$$

The Jacobian of $\tilde{f}$ at $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$:

$$\begin{aligned}A &= \begin{pmatrix} \frac{\partial \tilde{f}_1}{\partial \tilde{y}_1} & \frac{\partial \tilde{f}_1}{\partial \tilde{y}_2} \\ \frac{\partial \tilde{f}_2}{\partial \tilde{y}_1} & \frac{\partial \tilde{f}_2}{\partial \tilde{y}_2} \end{pmatrix} \bigg|_{\tilde{y}=\mathbf{0} \text{ (zero vector)}} = \begin{pmatrix} -b\tilde{y}_2 & -b\left(\tilde{y}_1 + \frac{l}{k}\right) \\ k\left(\tilde{y}_2 + \frac{a}{b}\right) & k\tilde{y}_1 \end{pmatrix} \bigg|_{\tilde{y}=\mathbf{0} \text{ (zero vector)}} \\ &= \begin{pmatrix} 0 & -\frac{l}{k}b \\ k\frac{a}{b} & 0 \end{pmatrix}\end{aligned}$$

$$\text{Tr}(A) = 0 \quad \det A = al > 0$$

So, the critical point is a stable center.

Let us solve the equation around the

critical point:

$$\left. \begin{aligned} \tilde{y}_1' &= -\frac{l}{k} b \tilde{y}_2 \\ \tilde{y}_2' &= k \frac{a}{b} \tilde{y}_1 \end{aligned} \right\}$$

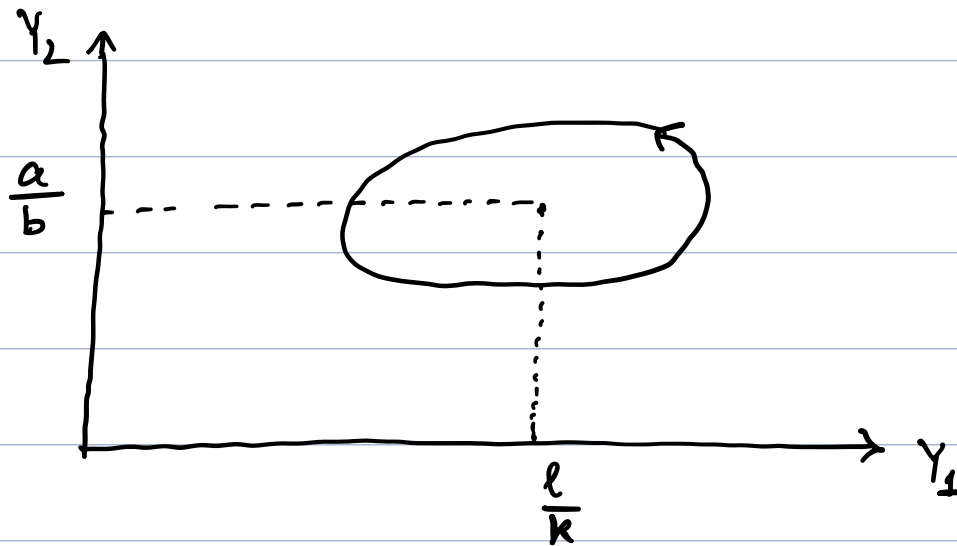
$$\Rightarrow k \frac{a}{b} \tilde{y}_1 \tilde{y}_1' = -\frac{l}{k} b \tilde{y}_2 \tilde{y}_2'$$

$$\Rightarrow k \frac{a}{b} \tilde{y}_1 \tilde{y}_1' = -\frac{l}{k} b \tilde{y}_2 \tilde{y}_2'$$

$$\Rightarrow k \frac{a}{2b} \tilde{y}_1^2 = -\frac{l}{2k} b \tilde{y}_2^2 + C$$

$$\Rightarrow \frac{ak}{b} \tilde{y}_1^2 + \frac{bl}{k} \tilde{y}_2^2 = C$$

$$\Rightarrow \boxed{\frac{ak}{b} \left(y_1 - \frac{l}{k} \right)^2 + \frac{bl}{k} \left(y_2 - \frac{a}{b} \right)^2 = C}$$



Transformation to a first order equation in the phase plane:

Consider a second order autonomous ODE

$$F(y, y', y'') = 0$$

We make a change of variables

$$\left. \begin{aligned} y_1 &= y \\ y_2 &= y_1' \end{aligned} \right\}$$

$$\text{and } y'' = y_2' = \frac{dy_2}{dt} = \frac{dy_2}{dy_1} \frac{dy_1}{dt} = \frac{dy_2}{dy_1} y_2$$

$$\text{The ODE becomes } \boxed{F\left(y_1, y_2, \frac{dy_2}{dy_1} y_2\right) = 0}$$

Ex: Free Undamped pendulum:

$$\theta'' + k \sin \theta = 0$$

Making the substitution:

$$\frac{dy_2}{dy_1} y_2 + k \sin(y_1) = 0$$

$$y_2 dy_2 = -k \sin(y_1) dy_1$$

$$\Rightarrow \boxed{\frac{y_2^2}{2} = k \cos(y_1) + C}$$